



Experiment No: 05

Name of the Experiment: The BJT Biasing Circuits.

Objective:

Study of the BJT Biasing Circuits.

Theory :

Unlike the diode, which has two doped region, a transistor has three doped region. They are as follows –

- a) Emitter,
- b) Base and
- c) Collector.

These three doped regions form two junctions: One between the emitter and base and other between the collector and the base. Because of these it can be thought as combination of two diodes, the emitter and the base form one diode and the collector and base form another diode. The emitter is heavily doped. Its job is to emit or inject free majority carrier (electron for NPN and hole for PNP) into the base. The base is lightly doped and very thin. It passes the most of the emitter-injected electron (for NPN) into the collector. The doping level of the collector is between emitter and base. Figure 5.1 shows the biased NPN transistor.

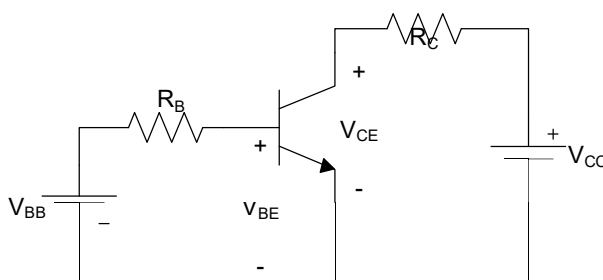


Figure 5.1 : Biasing of an NPN transistor.

If the V_{BB} is greater than the barrier potential, emitter electron will enter base region. The free electron can flow either into the base or into the collector. As base lightly doped and thin, most of the free electron will enter into the collector.

There are three different current in a transistor. They are emitter current (I_E), collector current (I_C) and the base current (I_B) are shows in figure 5.2.

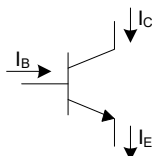


Figure 5.2 : Different current in transistor.



Here, $I_E = I_C + I_B$, and the current gain $\beta = \frac{I_C}{I_B}$

Characteristics Curve : The characteristics of a transistor is measured by two characteristics curve. They are as follows –

- Input characteristics curve.
- Output characteristics curve.

Input Characteristics Curve : Input characteristics is defined as the set of curves between input current (I_B) vs. input voltage (V_{BE}) for the constant output voltage (V_{CE}). It is the same curve that is found for a forward biased diode.

Output Characteristics Curve : Output characteristics is defined by the set of curves between output current (I_C) vs. output voltage (V_{CE}) for the constant input current (I_B). The curve has the following features –

- It has three regions namely Saturation, Active and Cutoff region.
- The rising part of the curve, where V_{CE} is between 0 and approximately 1 volt is called saturation region. In this region the collector diode is not reversed biased.
- When the collector diode of the transistor becomes reverse biased, the graph becomes horizontal. In this region the collector remains almost constant. This region is known as the active region. In applications where the transistor amplifies weak radio and TV signal, it will always be operation in the active region.
- When the base current is zero, but there is some collector current. This region of the transistor curve is known as the cutoff region. The small collector current is called collector cutoff current.
- For different value of base current (I_B) an individual curve can be obtained.

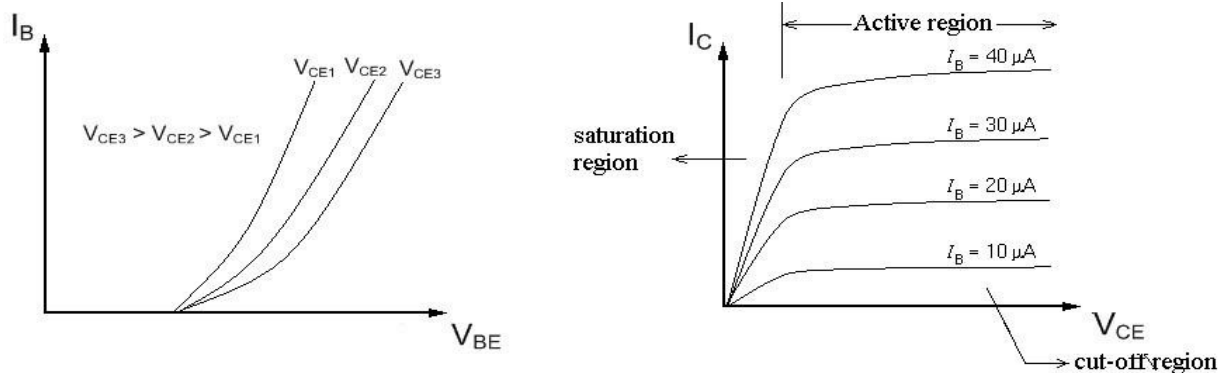


Figure 5.3 : (a) Input Characteristic, (b) Output Characteristic of NPN transistor.



Biassing a BJT circuit means to provide appropriate direct potentials and currents, using external sources, to establish an operating point or Q-point in the active region. Once the Q-point is established, the time varying excursions of input signal should cause an output signal of same waveform. If the output signal is not a faithful reproduction of the input signal, for example, if it is clipped on one side, the operating point is unsatisfactory and should be relocated on the collector characteristics. Therefore, the main objective of biasing a BJT circuit is to choose the proper Q-point for faithful reproduction of the input signal. There are different types of biasing circuit. However, in the laboratory, we will study only the fixed bias and self bias circuit. In the fixed bias circuit, shown if figure 5.4, the base current I_B is determined by the base resistance R_B and it remains constant. The main drawback of this circuit is the instability of Q-point with the variation of β of the transistor. In the laboratory, we will test the stability using two transistors with different β . In the self bias circuit shown if figure 5.5, this problem is overcome by using the self biasing resistor R_E to the emitter terminal.

Equipments And Components :

Serial no.	Component Details	Specification	Quantity
1.	NPN Transistor	C828, BD135	1 piece each
2.	Resistor	470 Ω , 560 Ω , 220K Ω , 10 K Ω	1 piece each
3.	POT	10K Ω , 100 K Ω	1 unit
4.	Bread Board		1 unit
5.	DC Power Supply		1 unit
6.	Digital Multimeter		1 unit
7.	Chords and wire		as required



Experimental Setup:

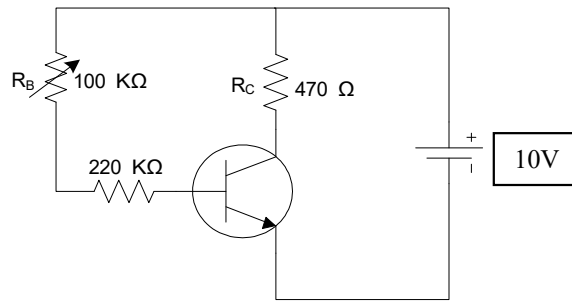


Figure 5.4: Experimental Circuit 1.

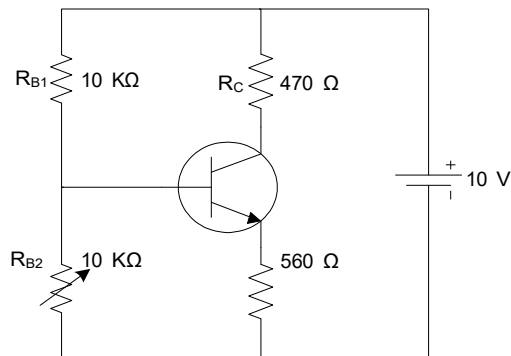


Figure 5.5: Experimental Circuit 2.

Procedure:

1. Arrange the circuit shown in figure – 5.4 by C828. Record R_C and set R_B to maximum value.
2. Decrease POT R_B gradually so that $V_{CE} = V_{CC} / 2$.
3. Measure voltage across R_C and V_{CE} .
4. Record the Q-point (V_{CE} , I_C).
5. Replace the C828 transistor by BD135 and repeat step 3 and 4.
6. Arrange the circuit shown in figure – 5.5 by C828. Record R_C and set R_B to minimum value.
7. Increase POT R_{B2} gradually so that $V_{CE} = V_{CC} / 2$.
8. Measure voltage across R_C and V_{CE} .
9. Record the Q-point (V_{CE} , I_C).
10. Replace the C828 transistor by BD135 and repeat step 8 and 9.



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Data Sheet :

Table 5.1 : Data for Fixed Bias Circuit.

Transistor	R_C (Ω)	V_C (volt)	$I_C = V_C / R_C$ (Amp)	V_{CE} (volt)	Q-point
C828					
BD135					

Table 5.2 : Data for Self Bias Circuit.

Transistor	R_C (Ω)	V_C (volt)	$I_C = V_C / R_C$ (Amp)	V_{CE} (volt)	Q-point
C828					
BD135					

Report:

1. Which circuit shows better stability? Explain in the context of the results obtained in the laboratory.
2. Draw the DC load line for both the circuits and show the Q-point.

Objective:

The objective of the experiment is BJT Biasing Circuits.

Theory:

Unlike the diode, which has two doped region, a transistor has three doped region. They are as follows -

a) Emitter

b) Base &

c) Collector.

These three doped regions form two junctions:

One between the emitter and base and other between the collector and the base. Because of these it can be thought as combination of two diodes, the emitter and the base form one diode and the collector and base form another diode. The emitter is heavily doped.

Its job is to emit or inject free majority carriers (electron for NPN and hole for PNP) into the base.

The base is lightly doped and very thin. It passes the most of the emitter-injected electron (for NPN) into the collector. The doping level of the collector is between emitter and base.

The free electron can flow either into the base or into the collector. As base is lightly doped and thin, most of the free electron will enter into the collector.

There are three different currents in a transistor. They are emitter current (I_E), collector current (I_C) and the base current (I_B).

Here,
$$I_E = I_B + I_C$$

and the current gain,
$$\beta = \frac{I_C}{I_B}$$

Characteristics Curve:

The characteristics of a transistor is measured by two characteristics curve. They are as follows -

a) Input characteristics curve.

b) Output characteristics curve.

Equipments & Components:-

(i) NPN Transistor (C828, BD135)

(ii) Resistor (470Ω , 560Ω , $220k\Omega$, $10k\Omega$)

(iii) POT ($10k\Omega$, $100k\Omega$)

(iv) Bread Board.

(v) DC Power Supply.

(vi) Digital Multimeter.

(vii) Cables & Wire.

Circuit Diagrams:-

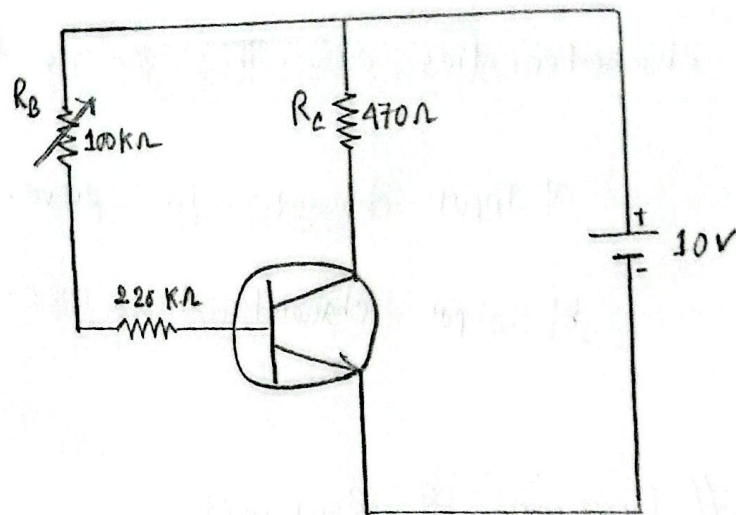


Fig: Experimental Circuit-01 (Fixed)

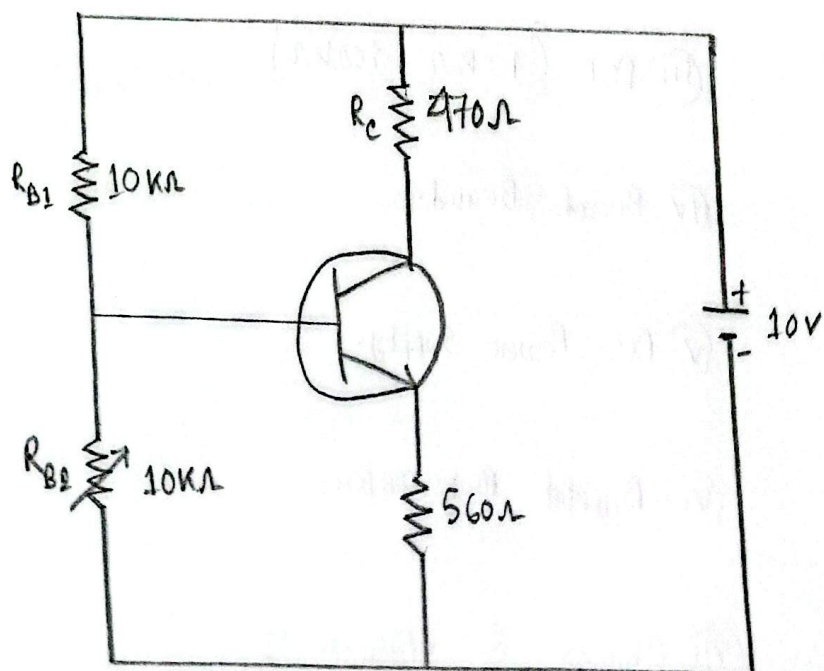


Fig: Experimental Circuit-02 (Self)

$I_{C(mA)}$

30
28
26
24
22
20
18
16
14
12
11
10
8
6
4
2
0

2

4

6

8

V_{CE}

• Fixed Bias Q-Point
• Self Bias Q-Point

Data Table:-

Table: 5.1 : Data for Fixed Bias Circuit:-

Transistor	$R_c (\Omega)$	$V_c (V_{OH})$	$I_c = V_c / R_c$ (Amp)	$V_{CE} (V_{OH})$	Q-Point
C828	462Ω	5.07 V	0.011 mA	5.00 V	(5.00, 0.011)
BD 135	462Ω	5.04 V	0.011 mA	5.02 V	(5.02, 0.011)

Table: 5.2: Data for Self Bias Circuit:-

Transistor	$R_c (\Omega)$	V_c (V _{OH})	$I_c = V_c / R_c$ (Amp)	V_{CE} (V _{OH})	Q-Point
C828	462Ω	4.55 V	9.84 mA	5.04 V	(5.04, 9.84 mA)
BD 135	462Ω	2.25 V	4.87 mA	5.05 V	(5.05, 4.87 mA)

Result Analysis :-

There is two types of circuit in this experiment. Now, these are the result analysis between fixed bias circuit and self bias circuit:

About temperature stability $\rightarrow$ Fixed bias have poor stability
And, self bias have good stability.

About Q-Point stability $\rightarrow$ Fixed bias have poor & low stability.
And, self bias have excellent stability.

About β dependence $\rightarrow$ Fixed bias have high independence.
And, self bias have low independence.

About Negative^{Feedback} $\rightarrow$ Fixed bias have no negative feedback.

And, self bias have negative feedback.

About transistor replacement $\rightarrow$ Fixed bias have large Q-point shift.

And, self bias have small Q-point shift.

About laboratory result $\rightarrow$ Fixed bias have less stable result.

And, self bias have more stable result.

Answer to the Question - 01:-

The self-Bias (Voltage Divider Bias) circuit shows better stability than the Fixed Bias circuit.

According to the observation table,

Fixed Bias:-

Two different transistors (C828 & BD135) produced noticeable differences in collector current and operating point.

Since Fixed Bias depends strongly on transistor β

(i) Replacing the transistor changes I_c ,

(ii) V_{CE} changes.

(iii) the Q-point shifts.

This indicates poor stability bias.

Self Bias:-

When the transistor was changed,

the operating point ~~remained~~ remained much closer to the desired operating region because the emitter resistor provided negative feedback.

Although, the collector current changed,

the circuit maintained nearly the same operating condition.

Hence,

- ① less Q -point movement.
- ② better thermal stability.
- ③ better transistor interchangeability.

The experimental results agree with transistor theory.

Therefore, the self bias (Voltage Divider Bias) circuit is more stable than the Fixed Bias circuit because it provides negative feedback, making the Q -point much less sensitive to transistor β and temperature variations.

Answer to

Answer to the Question - 02:-

For both circuits the collector circuit is essentially the same.

From the experiment,

Supply voltage, $V_{CC} = 10V$

Collector resistor, $R_C = 462\Omega$

Load line equation,

Using KVL,
 $V_{CC} = I_C R_C + V_{CE}$

$$\Rightarrow V_{CE} = 10 - 462 I_C$$

where, I_C is in amperes.

End points of the load line,

Cutoff, $I_C = 0 \text{ mA}$

$$V_{CE} = 10V$$

Point, $(10V, 0mA)$

Saturation,

$$V_{CE} = 0$$

$$I_C = \frac{10}{462}$$

$$I_C = 21.65 \text{ mA}$$

Point, $(0V, 21.65 \text{ mA})$

Approximate Q-points from the observation table:-

Fixed Bias:-

From the measurements,

$$V_{CE} \approx 5V$$

$$I_C \approx 1.1 \text{ mA}$$

Therefore, $Q_F = (5V, 1.1 \text{ mA})$

Self Bias:-

From the measurements, $V_{CE} \approx 5V$

$$I_C \approx 9.8 \text{ mA}$$

Therefore, $Q_S = (5V, 9.8 \text{ mA})$

Discussion:

The laboratory experiment demonstrates that the Self Bias (Voltage Divider Bias) circuit provides superior bias stability compared with the Fixed Bias circuit. The emitter resistor introduces negative feedback, reducing the influence of transistor gain (β), temperature changes, and device replacement on the operating point. The DC load line for both circuits extends from ($V_{CE} = 10V, I_C = 0 \text{ mA}$) to ($V_{CE} = 0V, I_C = 21.65 \text{ mA}$).

Based on the measured results, the Fixed Bias Q-point is approximately ($5V, 1.1 \text{ mA}$), while the Self-Bias Q-point is approximately ($5V, 9.8 \text{ mA}$).

These observations confirm that the self-bias circuit maintains a more consistent operating point and is therefore the preferred biasing method for practical transistor amplifier designs.



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Data Sheet : Group : 11

14/7

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